

YOLO Object and Lane Detection

1.0

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Chapter 1

Namespace Index

1.1 Namespace List

Here is a list of all namespaces with brief descriptions:

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Chapter 2

File Index

2.1 File List

Here is a list of all files with brief descriptions:

augment_dataset.py	11
train_yolo.py	11

Chapter 3

Namespace Documentation

3.1 augment_dataset Namespace Reference

Functions

- def [read_yolo_label](#) (label_path)
- def [write_yolo_label](#) (label_path, bboxes)
- def [augment_dataset](#) (input_dataset_dir, output_dataset_dir, num_augmentations=3, save_empty_labels=False)

Variables

- string [input_dataset_dir](#) = "./dataset_original"
- string [output_dataset_dir](#) = "./dataset"
- int [num_augmentations](#) = 3
- bool [save_empty_labels](#) = False

3.1.1 Function Documentation

3.1.1.1 augment_dataset()

```
def augment_dataset.augment_dataset (
    input_dataset_dir,
    output_dataset_dir,
    num_augmentations = 3,
    save_empty_labels = False )
```

Aplica augmentações ao dataset YOLO, gerando novas imagens e labels.

Não salva imagens sem bounding boxes válidas, a menos que save_empty_labels=True.

Args:

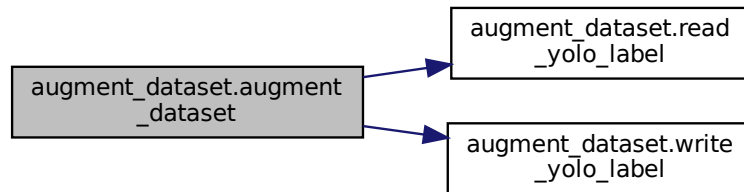
input_dataset_dir: Caminho do dataset original (ex.: yolov8_last/).

output_dataset_dir: Caminho onde o dataset aumentado será salvo.

num_augmentations: Número de imagens aumentadas por imagem original.

save_empty_labels: Se True, cria arquivos .txt vazios para imagens sem bboxes válidas.

Here is the call graph for this function:

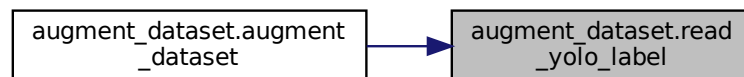


3.1.1.2 read_yolo_label()

```
def augment_dataset.read_yolo_label (
    label_path )
```

Lê um arquivo de anotações YOLO e retorna as bounding boxes.

Here is the caller graph for this function:

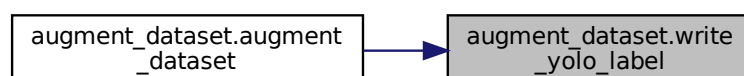


3.1.1.3 write_yolo_label()

```
def augment_dataset.write_yolo_label (
    label_path,
    bboxes )
```

Escreve as bounding boxes no formato YOLO em um arquivo.

Here is the caller graph for this function:



3.1.2 Variable Documentation

3.1.2.1 input_dataset_dir

```
string augment_dataset.input_dataset_dir = "./dataset_original"
```

3.1.2.2 num_augmentations

```
int augment_dataset.num_augmentations = 3
```

3.1.2.3 output_dataset_dir

```
string augment_dataset.output_dataset_dir = "./dataset"
```

3.1.2.4 save_empty_labels

```
bool augment_dataset.save_empty_labels = False
```

3.2 train_yolo Namespace Reference

Variables

- string [project_dir](#) = "./models"
- string [experiment_name](#) = "yolo_object_lane"
- [weights_dir](#) = os.path.join([project_dir](#), [experiment_name](#), "weights")
- [exist_ok](#)
- [model](#) = YOLO("yolo11n.pt")
- [results](#)
- [val_results](#)

3.2.1 Variable Documentation

3.2.1.1 exist_ok

```
train_yolo.exist_ok
```

3.2.1.2 experiment_name

```
string train_yolo.experiment_name = "yolo_object_lane"
```

3.2.1.3 model

```
train_yolo.model = YOLO("yolo11n.pt")
```

3.2.1.4 project_dir

```
string train_yolo.project_dir = "./models"
```

3.2.1.5 results

```
train_yolo.results
```

Initial value:

```
00001 = model.train(  
00002     data="./dataset/data.yaml", # Path to the dataset configuration file  
00003     epochs=200, # Total number of training epochs  
00004     imgsz=320, # Image size (optimized for Jetson)  
00005     hsv_h=0.015, # Hue adjustment for data augmentation  
00006     hsv_s=0.1, # Saturation adjustment for data augmentation  
00007     hsv_v=0.1, # Value (brightness) adjustment for data augmentation  
00008     translate=0.0, # Translation offset for data augmentation  
00009     scale=0.0, # Scale for data augmentation  
00010     flipplr=0.0, # Probability of horizontal flip  
00011     mosaic=0.0, # Probability of mosaic data augmentation  
00012     erasing=0.0, # Probability of random erasing  
00013     auto_augment=None, # Disable automatic augmentation for manual control  
00014     batch=8, # Batch size for training  
00015     amp=True, # Enable Mixed Precision to optimize GPU usage  
00016     device=0, # Use GPU 0 (or CPU if device=-1)  
00017     workers=2, # Number of workers for data loading  
00018     project=project_dir, # Base directory for saving results  
00019     name=experiment_name, # Experiment name for organization  
00020     exist_ok=True, # Allow overwriting previous results  
00021     freeze=0, # Do not freeze any model layers during training  
00022     lr0=0.001, # Initial learning rate  
00023     patience=0, # Disable early stopping  
00024     weight_decay=0.0005, # L2 regularization to prevent overfitting  
00025     save_period=20, # Save model checkpoints every 20 epochs  
00026     save=True, # Save the best model based on mAP metric  
00027 )
```

3.2.1.6 val_results

```
train_yolo.val_results
```

Initial value:

```
00001 = model.val(  
00002     data="./dataset/data.yaml", # Dataset configuration file for validation  
00003     imgsz=320, # Image size for validation  
00004     batch=16, # Batch size for validation (larger for powerful GPUs)  
00005     device='0' # Use GPU 0 for validation  
00006 )
```

3.2.1.7 weights_dir

```
train_yolo.weights_dir = os.path.join(project_dir, experiment_name, "weights")
```


Chapter 4

File Documentation

4.1 `augment_dataset.py` File Reference

Namespaces

- `augment_dataset`

Functions

- def `augment_dataset.read_yolo_label` (label_path)
- def `augment_dataset.write_yolo_label` (label_path, bboxes)
- def `augment_dataset.augment_dataset` (input_dataset_dir, output_dataset_dir, num_augmentations=3, save_empty_labels=False)

Variables

- string `augment_dataset.input_dataset_dir` = `"/dataset_original"`
- string `augment_dataset.output_dataset_dir` = `"/dataset"`
- int `augment_dataset.num_augmentations` = 3
- bool `augment_dataset.save_empty_labels` = False

4.2 `train_yolo.py` File Reference

Namespaces

- `train_yolo`

Variables

- string `train_yolo.project_dir` = `"/models"`
- string `train_yolo.experiment_name` = `"yolo_object_lane"`
- `train_yolo.weights_dir` = `os.path.join(project_dir, experiment_name, "weights")`
- `train_yolo.exist_ok`
- `train_yolo.model` = `YOLO("yolo11n.pt")`
- `train_yolo.results`
- `train_yolo.val_results`

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